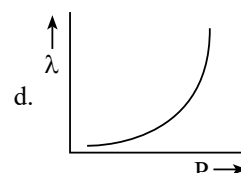
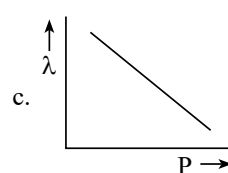
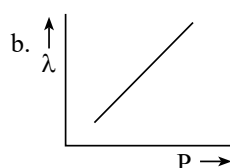
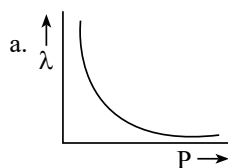


CHAPTER 11

Dual Nature of Radiation and Matter

Photoelectric Effect

- If ϕ is the work function of photosensitive material in eV and light of wavelength of numerical value $\lambda = \frac{hc}{e}$ metre, is incident on it with energy above its threshold value at an instant then the maximum kinetic energy of the photoelectron ejected by it at that instant (Take h -Planck's constant, c -velocity of light in free space) is (in SI units): (2024 Re)
 - $e + 2\phi$
 - $2e + \phi$
 - $e - \phi$
 - $e + \phi$
- The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons? (2023)
 - Na only
 - Cs only
 - Both Na and K
 - K only
- The maximum kinetic energy of the emitted photoelectrons in photoelectric effect is independent of: (2023-Manipur)
 - work function of material
 - intensity of incident radiation
 - frequency of incident radiation
 - wavelength of incident radiation
- The light rays having photons of energy 4.2 eV are falling on a metal surface having a work function of 2.2 eV. The stopping potential of the surface is: (2022 Re)
 - 6.4 V
 - 2 eV
 - 2 V
 - 1.1 V
- The threshold frequency of a photoelectric metal is ν_0 . If light of frequency $4\nu_0$ is incident on this metal, then the maximum kinetic energy of emitted electrons will be (2022 Re)
 - $4h\nu_0$
 - $h\nu_0$
 - $2h\nu_0$
 - $3h\nu_0$
- The graph which shows the variation of the de-Broglie wavelength (λ) of a particle and its associated momentum (P) is: (2022)



- The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be: ($h = 6.6 \times 10^{-34}$ Js) (2021)
 - 10^{17}
 - 10^{16}
 - 10^{15}
 - 10^{18}
- Light of frequency **1.5 times** the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled? (2020)
 - Four times
 - One-fourth
 - Zero
 - Doubled
- When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1 . When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2 . The ratio of v_1 to v_2 is: (2018)
 - 4 : 1
 - 1 : 4
 - 1 : 2
 - 2 : 1
- The photoelectric threshold wavelength of silver is 3250×10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is: (2017-Delhi)

(Given $h = 4.14 \times 10^{-15}$ eV and $c = 3 \times 10^8$ ms $^{-1}$)

 - $\approx 0.6 \times 10^6$ ms $^{-1}$
 - $\approx 61 \times 10^3$ ms $^{-1}$
 - $\approx 0.3 \times 10^6$ ms $^{-1}$
 - $\approx 6 \times 10^5$ ms $^{-1}$
- When a metallic surface is illuminated with radiation of wavelength λ , the stopping potential is V . If the same surface is illuminated with radiation of wavelength 2λ , the stopping potential is $V/4$. The threshold wavelength for the metallic surface is: (2016-I)
 - 4λ
 - 5λ
 - $\frac{5}{2}\lambda$
 - 3λ
- Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is: (2016-II)
 - 1 V
 - 3 V
 - +3 V
 - +4 V

13. A certain metallic surface is illuminated with monochromatic light of wavelength λ . The stopping potential for photoelectric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength for this surface for photoelectric effect is: (2015)

a. 4λ b. $\frac{\lambda}{4}$ c. $\frac{\lambda}{6}$ d. 6λ

14. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is:

(h = Planck's constant, c = speed of light) (2015 Re)

a. $\frac{hc}{3\lambda}$ b. $\frac{hc}{2\lambda}$ c. $\frac{hc}{\lambda}$ d. $\frac{2hc}{\lambda}$

15. When the energy of the incident radiation is increased by 20%, the kinetic energy of the photoelectrons emitted from a metal surface increased from 0.5 eV to 0.8 eV. The work function of the metal is: (2014)

a. 0.65 eV b. 1.0 eV c. 1.3 eV d. 1.5 eV

16. For photoelectric emission from certain metal the cut-off frequency is ν . If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be: (m is the electron mass) (2013)

a. $2\sqrt{\frac{h\nu}{m}}$ b. $\sqrt{\frac{h\nu}{2m}}$ c. $\sqrt{\frac{h\nu}{m}}$ d. $\sqrt{\frac{2h\nu}{m}}$

Unit Conversion

17. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly. (2020)

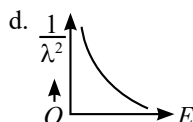
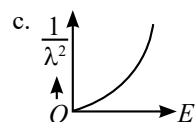
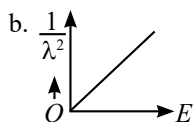
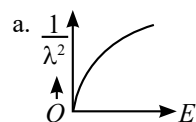
a. 0.6 b. 0.06 c. 0.006 d. 6

Wave Nature of Matter (De-Broglie Wavelength)

18. An electron and an alpha particle are accelerated by the same potential difference. Let λ_e and λ_α denote the de-Broglie wavelengths of the electron and the alpha particle, respectively, then: (2024 Re)

a. $\lambda_e > \lambda_\alpha$ b. $\lambda_e = 4\lambda_\alpha$ c. $\lambda_e = \lambda_\alpha$ d. $\lambda_e < \lambda_\alpha$

19. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle): (2024)



20. The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to: (2023)

a. V^2 b. $\sqrt{V}$ c. $\frac{1}{V}$ d. $\frac{1}{\sqrt{V}}$

21. The de Broglie wavelength associated with an electron, accelerated by a potential difference of 81 V is given by: (2023-Manipur)

a. 13.6 nm b. 136 nm c. 1.36 nm d. 0.136 nm

22. When two monochromatic light of frequency, ν and ν are incident on a photoelectric metal, their stopping potential becomes $V_s/2$ and V_s respectively. The threshold frequency for this metal is: (2022)

a. $\frac{3}{2}\nu$ b. 2ν c. 3ν d. $\frac{2}{3}\nu$

23. An electromagnetic wave of wavelength ' λ ' is incident on a photosensitive surface of **negligible** work function. If ' m ' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d , then: (2021)

a. $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$ b. $\lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$

c. $\lambda = \left(\frac{2h}{mc}\right)\lambda_d^2$ d. $\lambda = \left(\frac{2m}{hc}\right)\lambda_d^2$

24. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is: (2020)

a. 10^2 V b. 10^3 V c. 10^4 V d. 10 V

25. The de Broglie wavelength of an electron moving with kinetic energy of 144 eV is nearly, (2020-Covid)

a. 102×10^{-4} nm b. 102×10^{-5} nm
c. 102×10^{-2} nm d. 102×10^{-3} nm

26. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly) : ($m_e = 9 \times 10^{-31}$ kg) (2019)

a. 12.2×10^{-13} m b. 12.2×10^{-12} m
c. 12.2×10^{-14} m d. 12.2 nm

27. An electron of mass m with an initial velocity $\vec{V} = V_0 \hat{i}$ ($V_0 > 0$) enters an electric field $\vec{E} = -E_0 \hat{i}$ ($E_0 = \text{constant} > 0$) at $t = 0$. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is (2018)

a. $\lambda_0 t$ b. $\lambda_0 \left(1 + \frac{eE_0}{mV_0} t\right)$
c. $\frac{\lambda_0}{\left(1 + \frac{eE_0}{mV_0} t\right)}$ d. λ_0

28. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m , is: (2017-Delhi)

a. $\frac{h}{\sqrt{3mkT}}$ b. $\frac{2h}{\sqrt{3mkT}}$ c. $\frac{2h}{\sqrt{mkT}}$ d. $\frac{h}{\sqrt{mkT}}$

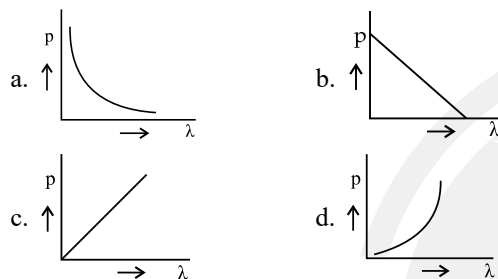
29. An electron of mass m and a photon have same energy E . The ratio of de-Broglie wavelengths associated with them is (c being velocity of light) (2016-I)

a. $\frac{1}{c} \left(\frac{E}{2m} \right)^{\frac{1}{2}}$ b. $\left(\frac{E}{2m} \right)^{\frac{1}{2}}$ c. $c(2mE)^{\frac{1}{2}}$ d. $\frac{1}{c} \left(\frac{2m}{E} \right)^{\frac{1}{2}}$

30. Electrons of mass m with de-Broglie wavelength λ fall on the target in an X-ray tube. The cutoff wavelength (λ_0) of the emitted X-ray is: (2016-II)

a. $\lambda_0 = \frac{2m^2 c^2 \lambda^2}{h^2}$ b. $\lambda_0 = \lambda$
c. $\lambda_0 = \frac{2mc\lambda^2}{h}$ d. $\lambda_0 = \frac{2h}{mc}$

31. Which of the following figures represent the variation of particle momentum and the associated de-Broglie wavelength? (2015)



32. If the kinetic energy of the particle is increased to 16 times its previous value, the percentage change in the de-Broglie wavelength of the particle is (2014)

a. 25 b. 75 c. 60 d. 50

33. The wavelength λ_e of an electron and λ_p of a photon of same energy E are related by: (2013)

a. $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$ b. $\lambda_p \propto \lambda_e^2$
c. $\lambda_p \propto \lambda_e$ d. $\lambda_p \propto \sqrt{\lambda_e}$

Parameters of Photon (Momentum, Pressure and Energy)

34. If c is the velocity of light in free space, the correct statements about photon among the following are : (2024)

- A. The energy of a photon is $E = hv$
B. The velocity of a photon is c .
C. The momentum of a photon, $p = \frac{hv}{c}$
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.

Choose the correct answer from the options given below:

- a. A, C and D only b. A, B, D and E only
c. A and B only d. A, B, C and D only

35. Light of wavelength 5000 nm is incident on a metal with work function 2.28 eV. The de-Broglie wavelength of the emitted electron is: (2015 Re)

a. $\leq 2.8 \times 10^{-12} \text{ m}$ b. $< 2.8 \times 10^{-10} \text{ m}$
c. $< 2.8 \times 10^{-9} \text{ m}$ d. $\geq 2.8 \times 10^{-9} \text{ m}$

Davisson and Germer Experiment

36. The wave nature of electrons was experimentally verified by. (2020-Covid)

- a. Hertz b. Einstein
c. Davisson and Germer d. de-Broglie

Answer Key

1. (c) 2. (b) 3. (b) 4. (c) 5. (d) 6. (a) 7. (b) 8. (c) 9. (c) 10. (a, d)
11. (d) 12. (b) 13. (a) 14. (b) 15. (b) 16. (d) 17. (b) 18. (a) 19. (b) 20. (c)
21. (d) 22. (None) 23. (b) 24. (c) 25. (d) 26. (b) 27. (c) 28. (a) 29. (a) 30. (c)
31. (a) 32. (b) 33. (b) 34. (d) 35. (d) 36. (c)